

Frontera eficiente

Dado un retorno objetivo específico r^* , resolver

$$\min_w w^T \Sigma w$$

$$\text{Sujeto a } w^T \mu = r^*, \quad 1^T w = 1$$

1: Ecuación inicial

$$f(w) = w^T \Sigma w$$

3: Construcción del Lagrangiano

$$f(w, \lambda_1, \lambda_2) = w^T \Sigma w - \lambda_1 (w^T \mu - r^*) - \lambda_2 (1^T w - 1)$$

$$f(w, \lambda_1, \lambda_2) = w^T \Sigma w - \lambda_1 w^T \mu - \lambda_1 r^* - \lambda_2 1^T w - \lambda_2$$

2: Restricciones

$$g(w) = w^T \mu = r^*$$

$$h(w) = 1^T w = 1$$

5: Aislar el vector incógnita

$$2 \Sigma w - \lambda_1 \mu - \lambda_2 1 = 0$$

$$2 \Sigma w = \lambda_1 \mu + \lambda_2 1$$

$$\Sigma w = \frac{\lambda_1}{2} \mu + \frac{\lambda_2}{2} 1$$

4: Derivadas matriciales e igualar a 0

$$\frac{\partial L}{\partial w} = 2 \Sigma w - \lambda_1 \mu - \lambda_2 1 = 0$$

$$\frac{\partial L}{\partial \lambda_1} = w^T \mu - r^* = 0$$

$$\frac{\partial L}{\partial \lambda_2} = 1^T w - 1 = 0$$

Si

$$\gamma = \begin{pmatrix} \lambda_1 / 2 \\ \lambda_2 / 2 \end{pmatrix}, \quad A = \begin{pmatrix} \mu & 1 \end{pmatrix}$$

$$\Sigma w = A \gamma$$

$$w = \Sigma^{-1} A \gamma$$

6: Juntar restricciones y reemplazar

$$1^T w = 1, \quad w^T \mu = r^*$$

Como son vectores del mismo tamaño $a^T b = b^T a$

$$1^T w = 1 \quad \mu^T w = r^*$$

Recordando que 1^T y μ^T son vectores $1 \times n$

$$\begin{pmatrix} 1 & 1 & 1 \\ \mu_1 & \mu_2 & \mu_3 \end{pmatrix} = A^T = \begin{pmatrix} 1^T \\ \mu^T \end{pmatrix}$$

$$A^T w = \begin{pmatrix} 1 \\ r^* \end{pmatrix}$$